

TAEKYUNG KI

[Homepage](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#) | E-mail: taekyung.ki@kaist.ac.kr

RESEARCH INTERESTS

I am interested in generative models and their applications to creating real-world impact for human. My current research focuses on **controllable, interactive, and reliable video generation** toward **the broader goal of video-based world models**. My previous work spans human video generation, including audio-driven avatar generation for video dubbing [C2], diverse real-time controllability [C4], user-interactive head avatar generation [C6], NeRF-based parametric head reenactment [C3], and training-free methods for controllable and reliable video diffusion sampling [C5, C7].

EDUCATION

Ph.D. in Artificial Intelligence, KAIST

- Advisor: [Prof. Sung Ju Hwang](#)

Sep. 2025 - Present

Seoul, South Korea

M.S. in Mathematics, Yonsei University

- Thesis: Deep Scattering Transform with Max-pooling

- Advisor: [Prof. Youngmi Hur](#)

Mar. 2019 - Feb. 2021

Seoul, South Korea

B.S. in Mathematics, Kookmin University

- Ranked 1st in department of mathematics.

- Advisor: [Prof. Won-Kwang Park](#)

Mar. 2015 - Feb. 2019

Seoul, South Korea

PUBLICATIONS

*: Equal contribution / †: Corresponding author / C: Conference publication / W: Workshop publication / P: Preprint

[P1] **HINT-SD: Targeted Hindsight Self-Distillation for Long-Horizon Agents**

Woongyeng Yeo*, Yumin Choi*, [Taekyung Ki](#), and Sung Ju Hwang

arXiv 2026

[C7] **Self-Refining Video Sampling**

Sangwon Jang*, [Taekyung Ki](#)*, Jaehyeong Jo*, Saining Xie, Jaehong Yoon†, and Sung Ju Hwang†

International Conference on Machine Learning (ICML) 2026

[C6] **Avatar Forcing: Real-Time Interactive Head Avatar Generation for Natural Conversation**

[Taekyung Ki](#)*, Sangwon Jang*, Jaehyeong Jo, Jaehong Yoon, and Sung Ju Hwang†

Conference on Computer Vision and Pattern Recognition (CVPR) 2026

[C5] **Frame Guidance: Training-Free Guidance for Frame-Level Control in Video Diffusion Model**

Sangwon Jang*, [Taekyung Ki](#)*, Jaehyeong Jo, Jaehong Yoon, Soo Ye Kim, Zhe Lin, and Sung Ju Hwang†

International Conference on Learning Representations (ICLR) 2026

[C4] **FLOAT: Generative Motion Latent Flow Matching for Audio-driven Talking Portrait**

[Taekyung Ki](#), Dongchan Min, and Gyeongsu Chae

International Conference on Computer Vision (ICCV) 2025

Audio-Visual Generation & Learning Workshop at ICCV (AVGenL @ ICCV), 2025

- [C3] **Learning to Generate Conditional Tri-plane for 3D-aware Expression Controllable Portrait Animation**
Taekyung Ki, Dongchan Min, and Gyeongsu Chae
 European Conference on Computer Vision (**ECCV**) 2024
- [C2] **StyleLipSync: Style-based Personalized Lip-sync Video Generation**
Taekyung Ki^{*} and Dongchan Min^{*}
 International Conference on Computer Vision (**ICCV**) 2023
- [C1] **Deep Scattering Network with Max-pooling**
Taekyung Ki and Youngmi Hur[†]
 IEEE Data Compression Conference (**DCC**) 2021

EXPERIENCE

Apple ML Research Intern, Video computer vision (VCV) Team Research on 3D video diffusion models	<i>June 2026 - Sep. 2026</i> Sunnyvale, CA, US
KAIST Visiting Researcher, MLAI Lab. Research on video diffusion models (Advisor: Prof. Sung Ju Hwang)	<i>Mar. 2025 - Aug. 2025</i> Seoul, South Korea
DeepBrain AI Inc. Research Scientist (Alternative Military Service, South Korea) Developing novel frameworks for human video generation	<i>Dec. 2023 - Mar. 2025</i> Seoul, South Korea
AITRICS Research Scientist (Alternative Military Service, South Korea) Working with Prof. Sung Ju Hwang and Prof. Eunho Yang Developing novel frameworks for human video generation	<i>Mar. 2022 - Dec. 2023</i> Seoul, South Korea
JENTI Inc. Research Engineer OCR and NLP-based document analysis	<i>Feb. 2021 - Dec. 2021</i> Seoul, South Korea
Wavelets and Approximation Lab. at Yonsei University Graduate Student Researcher (Advisor: Prof. Youngmi Hur) Mathematical analysis on deep neural networks	<i>Aug. 2019 - Feb. 2021</i> Seoul, South Korea
Scientific Computing Lab. at Kookmin University Undergraduate Student Researcher (Advisor: Prof. Won-Kwang Park) Analysis (PDEs)	<i>Mar. 2018 - Feb. 2019</i> Seoul, South Korea

TALKS

Motion Latent Flow Matching for Real-time Audio-driven Talking Portrait at Pika Labs (Host: Chenlin Meng , CTO)	<i>Aug. 2025</i> Remote
---	----------------------------

TEACHING

Teaching Assistant at Yonsei University Calculus 1-2, Linear algebra, Mathematics and biology, and Engineering mathematics (Vector calculus, Fourier analysis, Linear PDEs and statistics, etc.)	<i>Mar. 2019 - Dec. 2020</i>
--	------------------------------

AWARDS AND SCHOLARSHIPS

1st Prize Winner at AI Grand Challenge 2021 , Ministry of Science and ICT	<i>Oct. 2021</i>
BK 21 Scholarship , Department of Mathematics, Yonsei University	<i>Sep. 2020 - Feb. 2021</i>
Top Honors in Mathematics , Department of Mathematics, Kookmin University	<i>Feb. 2019</i>
Academic Scholarship , Department of Mathematics, Kookmin University	<i>Sep. 2015 - Feb. 2019</i>

ACADEMIC SERVICES

Conference Reviewer

Neural Information Processing Systems (NeurIPS)	<i>2026 -</i>
Computer Vision and Pattern Recognition (CVPR)	<i>2025 -</i>
International Conference on Computer Vision (ICCV)	<i>2025 -</i>
European Conference on Computer Vision (ECCV)	<i>2026 -</i>
ACM SIGGRAPH Asia	<i>2026 -</i>
British Machine Vision Conference (BMVC)	<i>2026 -</i>
Association for Computational Linguistics Rolling Review (ACL ARR)	<i>2026 -</i>

REFERENCES

Prof. Sung Ju Hwang Associate Professor, Graduate School of AI, KAIST
- E-mail: sungju.hwang@kaist.ac.kr

Prof. Youngmi Hur Full Professor, Department of Mathematics, Yonsei University
- E-mail: yhur@yonsei.ac.kr

Prof. Won-Kwang Park Full Professor, Department of Mathematics, Kookmin University
- E-mail: parkwk@kookmin.ac.kr